

CSC1005/CSC1022 Database Lab

Lab 5: SQL Joins and ANY/ALL Solutions

Goal: Learn SQL joins syntax and how to write queries using ANY / ALL.

This lab is designed to get students familiar with using multiple tables and joins to retrieve results.

You have finished the lab when you have finished all queries.

If queries specify using a join, you must use a join

Section 1. Basketball (Make default database)

1. Find the NBA Coach of the Year IN 1995 and the number of games won

Ans:

```
SELECT c.coachID,c.won,c.year FROM coaches c,awards_coaches ac
WHERE c.coachID = ac.coachID
AND c.year = ac.year AND ac.year = 1995
AND ac.award = 'NBA Coach of the Year';
```

Students may also use Inner Join

CoachID	Won	Year
jacksph01	72	1995

2. Return the id of the player who in 1997 scored more than 1000 points and in 1996 played for Chicago Bulls and got more than 350 rebounds. **(Difficulty : Hard)**

(Hint: You could need to join the same table twice. You need to obtain the players' ID list matching the requirements for 1997, and the list for 1996. The inner join will give you the final player. You need to join again with another table to ensure the correct name of the team.)

Ans:

```
SELECT pt1.playerID, pt1.points,
FROM players_teams pt1, teams t, players_teams pt WHERE
pt1.points > 1000
and pt1.year = 1997 and pt1.playerID = pt.playerID
and t.tmID = pt.tmID and t.year = pt.year
and t.year = 1996 and t.name = 'chicago bulls'
and pt.rebounds > 350;
```

playerID	points
jordami01	2357

Section 2: Sakila (Make Sakila the default database)

3. List customers (customer id) whose total payments are less than at least one other customer's total payments. Order by Total payments (Hint: Use **payment table**)

Ans:

```
SELECT customer_id, SUM(amount) FROM payment
GROUP BY customer_id
HAVING SUM(amount) < SOME
(SELECT SUM(amount) FROM payment
GROUP BY customer_id)
order by SUM(amount);
```

customer_id	SUM(amount)
248	50.85
281	50.86
...	...

4. Return the actors full name in any movie containing the word 'pirates' in the title.

Ans:

```
Select a.first_name, a.last_name, fa.actor_id, f.title
from actor a, film_actor fa, film f
where fa.film_id = f.film_id
and f.title like '%pirates%'
and fa.actor_id = a.actor_id;
```

first_name	last_name	actor_id	f.title
HUMPHREY	WILLIS	164	PIRATES ROXANNE
NATALIE	HOPKINS	50	TWISTED PIRATES
JUDE	CRUISE	57	TWISTED PIRATES
ADAM	GRANT	71	TWISTED PIRATES

5. Return the count and the last name of the actor starring in the most number of films
(Difficulty : Hard)
(Hint: You may have to use nested query and group by and ANY / ALL keywords)

Ans:

```
SELECT a.last_name, count(a.last_name) as count
FROM actor a, film_actor fa
WHERE a.actor_id = fa.actor_id
GROUP BY a.last_name
HAVING count >= ALL (SELECT count(a.last_name)
FROM actor a, film_actor fa
WHERE a.actor_id = fa.actor_id
GROUP BY a.last_name)
```

last_name	count
KILMER	134

Section 3. World (Make default database)

6. Select countries whose population is greater than every country in the 'Caribbean' region. Display country name and the population. (Use ANY / ALL / SOME keyword)

Ans:

```
select Name, Population from country
WHERE population > (SELECT max(population) FROM country
WHERE region = 'Caribbean');
```

7. **Using JOIN**, get the name and the population of the Irish city with more than 200,000 inhabitants. (you can also practice using a nested query)

Ans:

```
select c.name, c.Population
from city c, country co
where c.CountryCode = co.Code
and co.Name = 'Ireland'
and c.Population > 200000;
```

name	Population
Dublin	481854

8. Return the name of the countries where English is the official language, the form of government is Republic and life expectancy is greater than 70 years. (Use Join)
(Hint: You need to join two tables)

Ans:

```
select c.Name, c.LifeExpectancy, c.GovernmentForm
from country c, countrylanguage l
where c.Code = l.CountryCode
and l.Language = 'English' and l.IsOfficial = 'T'
and c.GovernmentForm = 'Republic'
and c.LifeExpectancy > 70;
```

Name,	LifeExpectancy	GovernmentForm
Ireland	76.8	Republic
Malta	77.9	Republic
Seychelles	70.4	Republic